

Danni's 2026 Pedagogical Philosophy

The purpose of education is to **develop human capacity for independent judgment.**

Education should build the knowledge, mental models, critical reasoning, and metacognitive abilities that allow people to understand complex problems, evaluate evidence, challenge authoritative answers, revise their beliefs, and act responsibly. It should also teach learners to extend their capabilities through collaboration with other people and with machines.

- Education therefore has both an **individual and a collective responsibility**: to develop capable people today while cultivating the conditions through which capable people can continue to develop tomorrow.
- Technology should remove unnecessary barriers to learning, but it should not remove the cognitive work through which learners become capable of thinking without it.
- The defining educational question of the Ai-augmented era: **Does this system produce people capable of knowing, judging, troubleshooting, and advancing the field when existing tools, models, and assumptions fail?**

We already adopt many of these practices in our scientific training pipeline, which functions in many respects as a **knowledge-work apprenticeship**: trainees learn by working alongside more experienced scientists, observing good approaches to solving problems, receiving feedback and mentorship, collaborating with others, and having repeated opportunities to attempt, fail, and improve in relatively low-consequence environments. This model has deep roots in the traditions of higher education and scientific practice and has much in common with apprenticeship models used to develop expertise in other forms of skilled work. The visualization below, developed with the help of OpenAI, is my attempt to make the implicit capabilities developed through good scientific training explicit: to atomize the apprenticeship process into its component skills and consider how those skills might be deliberately developed across the broader knowledge-workforce in an Ai-rich world.



1. When output can be produced without learning, output is no longer sufficient evidence of learning.

Historically, we have often treated a finished product: a correct analysis, an essay, a program, a diagnosis, a solution, as a proxy for the learning that produced it. This has had the immense benefit of scalability for instructors. However, generative Ai weakens that assumption. When a learner can produce a sophisticated final product while bypassing much of the cognitive work required to create it, **the quality of the output no longer reliably tells us what the learner knows or can do independently.**

Assessment must therefore shift attention toward the *process of knowing*: the intermediate reasoning, decisions, representations, revisions, and judgments that occur on the way to a final answer. Therefore, we should increasingly assess the **unfinished milestones of thought**:

How was the problem framed? How was the final result checked?	What assumptions were identified? What evidence was considered or rejected? What alternatives were considered? What changed the learner's mind?
Why was a particular method chosen?	Where did uncertainty enter?

The goal is to make the learner's reasoning sufficiently visible so that we can distinguish *performance produced with assistance from capability that has actually been developed.*

2. Tool fluency without epistemic judgment is not education.

A learner producing a correct statistical analysis, Python program, diagnosis, scientific explanation, or Ai-generated synthesis is not necessarily evidence that education has succeeded. The more important question is whether the learner has developed enough conceptual structure and judgment to:

- **understand** why a result is reasonable;
- **recognize** when it is wrong;
- **identify** the assumptions on which it depends;
- **transfer** the underlying reasoning to a new situation;
- **determine** what evidence would change the conclusion; and
- **challenge** an apparently authoritative answer.

This distinction becomes increasingly important as tools become more capable. A result can be technically polished, internally coherent, and entirely plausible *while still being wrong because of a subtle problem in the data, an inappropriate model, an unstated assumption, a mismatch between the question and the method, or a failure of domain knowledge.*

Substantive validation therefore requires independent expertise. The educated learner must be able to move beyond asking “Does this output look right?” toward asking “What would have to be true for this conclusion to be right, and do I have sufficient evidence to believe those conditions hold?”

3. Educational optimization and productivity optimization are not the same objective.

In professional settings, efficiency is often desirable. In educational settings, some forms of efficiency *can undermine the very capability* we are trying to develop. Expertise develops through scaffolded encounters with progressively difficult problems. Learners attempt, struggle, make errors, receive feedback, revise their reasoning, compare alternatives, and reconcile their conclusions with evidence.

That cognitive work is not simply an inefficiency that better technology should eliminate. **In many cases, it is the mechanism through which learning occurs.** Education therefore requires **productive cognitive friction**: difficulty that serves a developmental purpose.

This creates an important distinction in the use of Ai: For an experienced practitioner who already possesses relevant expertise, asking Ai to generate a first draft, write routine code, summarize information, or automate familiar work may be a sensible productivity strategy. Ai can extend existing capability and scaffold cognition. For a novice who is still developing that expertise, having Ai perform the same cognitive task may remove precisely the reasoning episode the learner needs to experience. The principle is therefore not that learners should avoid Ai or that difficulty is inherently virtuous. **Ai should not substitute for the cognition that a learning experience is specifically intended to develop.**

The educational question becomes: **Which parts of this task should technology make easier, and which parts must the learner still do because doing them is how the capability is built?**

4. An educated person has intellectual agency.

An educated person is not simply someone who possesses information or can produce high-quality outputs. They possess **intellectual agency**: the capacity to form and revise judgments for themselves. They can frame problems, interrogate evidence, identify assumptions, audit reasoning, tolerate uncertainty, change their minds when the evidence warrants it, and recognize when an authority is wrong. That authority may be a professor, a paper, an institution, an expert, an algorithm, a statistical model, or an equation.

Mathematical and computational systems can create a particularly powerful illusion of objectivity. An educated person understands that algorithms and equations are created within systems of assumptions, measurement choices, objectives, constraints, and values. They therefore ask not only whether a calculation was performed correctly, but whether the model deserves to be applied to the problem in the first place.

The purpose of education is not to produce reflexive skepticism toward expertise. It is to develop people capable of determining **when trust is warranted, when scrutiny is required, and when established reasoning should be challenged.**

5. Education must preserve the cognitive commons.

Intellectual agency is not only an individual good. Professions depend on a shared **cognitive common**: the accumulated reservoir of expertise, judgment, tacit knowledge, problem-solving

ability, and institutional memory that allows a field to advance, troubleshoot failures, respond to novel situations, and correct itself.

Experienced experts do not simply appear. They exist because previous generations created opportunities for novices to become experts through accumulated, progressively demanding experiences: observation, practice, mentorship, failure, feedback, increasing responsibility, and repeated encounters with problems whose solutions were not already known.

Every organization benefits from this shared reservoir of expertise, including organizations that invest relatively little in developing novices themselves. Ai creates a potential collective-action problem. For any individual organization, it may be economically rational to automate entry-level work, reduce apprenticeship opportunities, or eliminate developmental positions while assuming that experienced experts will remain available to hire when needed. But if many organizations make the same decision, the developmental pathways through which future experts are produced may begin to disappear.

The result could be a system that becomes increasingly good at **producing answers while becoming progressively worse at producing people capable of independently evaluating those answers**. *Education therefore has an intergenerational responsibility.*